

Unit II

Research Design

Research Design

- Task of defining the research problem is the preparation of the research project, popularly known as the “research design”.
- Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study constitute a research design.
- A plan or strategy for conducting the research.
- A research design is one that minimizes bias and maximizes the reliability of the data.
- It also yields maximum information, gives minimum experimental error, and provides different aspects of a single problem.
- A research design depends on the purpose and nature of the research problem. Thus, one single design cannot be used to solve all types of research problem, i.e., a particular design is suitable for a particular problem.

Research Design

- What is the study about?
- Why is the study being made?
- Where will the study be carried out?
- What type of data is required?
- Where can the required data be found?
- What periods of time will the study include?
- What will be the sample design?
- What techniques of data collection will be used?
- How will the data be analyzed?
- In what style will the report be prepared?

Research Design

- Sampling Design: Method of selecting items to be observed for the given study
- Observational Design: Conditions under which observation must be made
- Statistical Design: How many data items to be observed? How information / data gathered has to be analysed?
- Operational Design: Techniques by which procedures specified in sampling, statistical and observational designs can be carried out.

Need of research design

- Smooth sailing of research operations.
- Stands for advance planning of the methods.
- Great bearing on reliability of the results.
- Helps to give directions.
- Helps in decision making.
- Research design prevent blind searching.
- Helps researchers to anticipate potential problems in collecting data etc.

Features of a Good Design

- ❖ The means of obtaining information
- ❖ The availability and skills of the researcher and his staff, if any;
- ❖ The availability of time and money for the research work.
- ❖ It should be flexible enough to consider different aspects of the study in case of exploratory.
- ❖ The design should be accurate with minimum bias in case of accurate description
- ❖ Control of extraneous variables
- ❖ Statistical correctness for testing hypothesis

Features of a Good Design

- Research design involves consideration of following factors:
 - Means of obtaining information
 - Availability and skills of researcher and his staff
 - Objective of the problem to be studied
 - Nature of the problem to be studied
 - Availability of time and money
- Exploratory / Formulative -> Research design must be flexible to permit consideration of many aspects of the phenomenon
- Hypothesis > Minimization of bias and maximization of reliability.

Concepts relating to research design

☐ Variables

- Dependent variables and independent variables
- Extraneous variables and confounded variables

☐ Research Hypothesis

- Null and Alternate hypothesis
- Directional and Non directional hypothesis

☐ Experimental and Control groups

☐ Treatments

Term	Role	Example
Dependent Variable (DV)	Measured outcome	Model accuracy (%)
Independent Variable (IV)	Factor manipulated	Training dataset size
Extraneous Variable	Uncontrolled factor, may affect DV	Learning rate, data quality
Confounded Variable	Extraneous variable linked to IV	Using advanced model for larger dataset

Variables

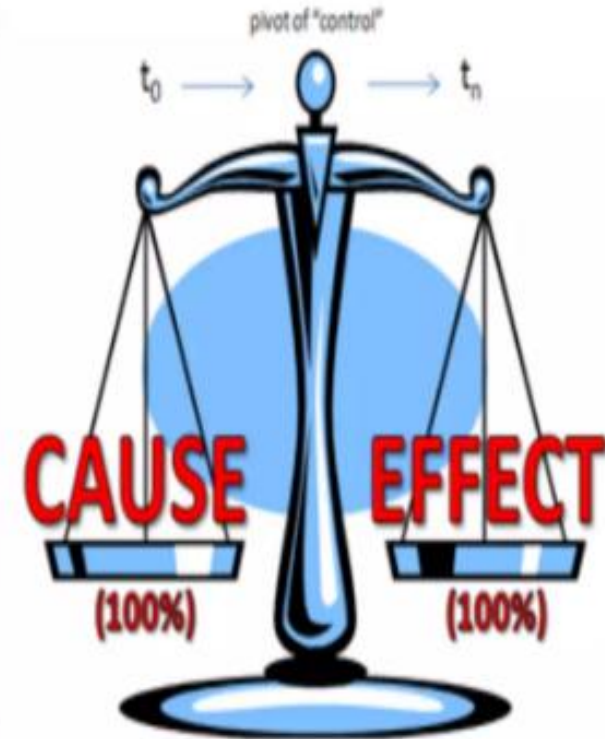
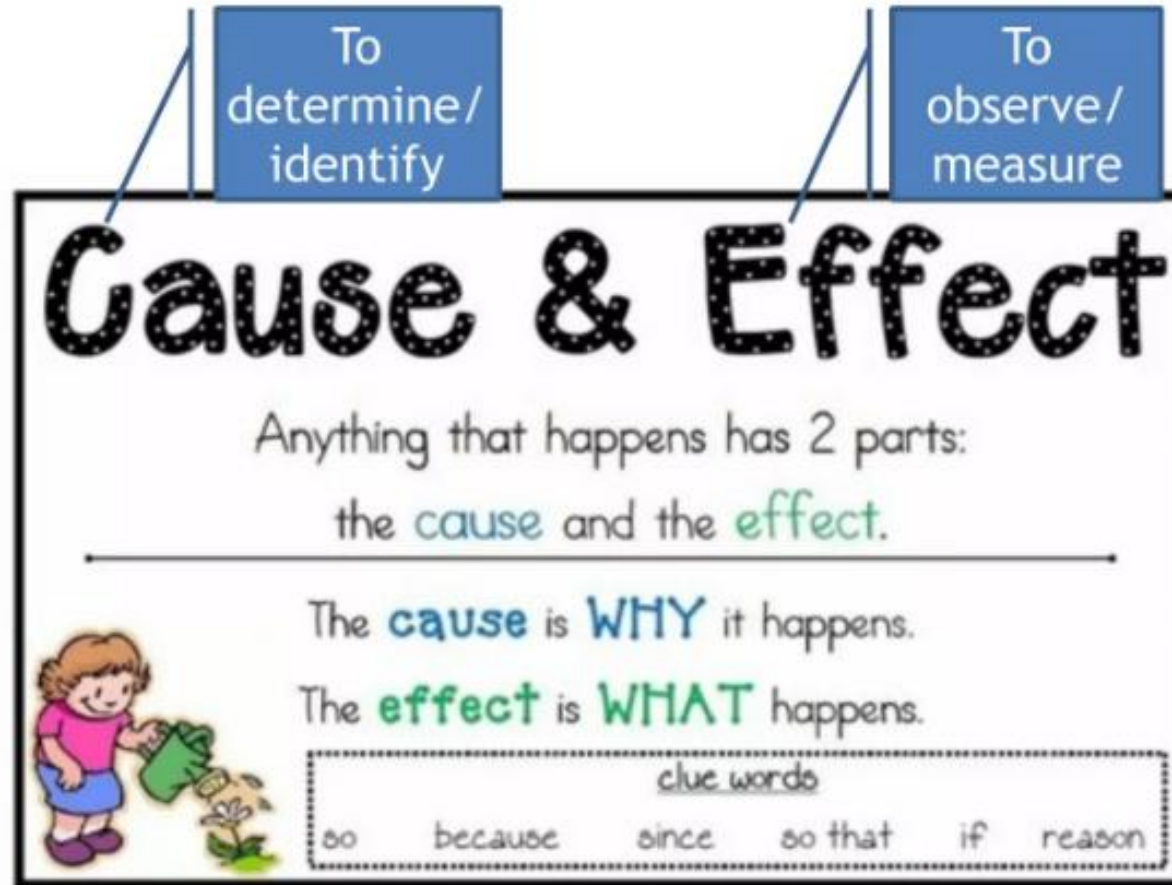
Any characteristics which is subject to change and can have more than one value. They play a crucial role in formulating hypotheses, designing experiments, and interpreting results.

Example: intelligence, motivation, age, gender

Quantitative phenomena characterized by presence / absence of attributes

Continuous and discrete variables

Concept of cause and effect



Types of Variables

Independent Variables

Independent variables are the factors that researchers manipulate or control in an experiment to observe their effect on the dependent variable. They are considered the "cause" in a cause-and-effect relationship.

Types:

1. Experimental Independent Variables: These are directly manipulated by the researcher (e.g., dosage of a drug).
2. Subject Variables: These are characteristics of the subjects that cannot be manipulated but can categorize participants (e.g., age, gender)

Dependent Variables

Dependent variables are the outcomes or effects that researchers measure in response to changes in the independent variable. They are considered the "effect" in a cause-and-effect relationship.

Characteristics: The values of dependent variables depend on the variations of independent variables.

- **Examples** include test scores, growth measurements, or behavioral responses

Extraneous Variables

Extraneous variables are any variables other than the independent variable that could influence the dependent variable. These can introduce noise into the experiment and potentially confound results if not controlled.

Examples: Environmental factors (temperature, lighting)

- Participant characteristics (mood, fatigue)
- Experimental conditions (time of day when measurements are taken)

Whatever effect is noticed on dependent variable as a result of extraneous variable(s) is technically described as an 'experimental error'. A study must always be so designed that the effect upon the dependent variable is attributed entirely to the independent variable(s), and not to some extraneous variable or variables

Confounded Variables

Confounded variables occur when two or more variables are intertwined in such a way that their individual effects on the dependent variable cannot be separated. This can lead to misleading conclusions about relationships between variables.

Characteristics: Confounding occurs when an extraneous variable correlates with both the independent and dependent variables.

Example include a study examines the effect of a new teaching method on student performance but does not control for prior knowledge, then prior knowledge becomes a confounding variable.

Importance of Variables in Research

Understanding variables is essential for several reasons:

- **Hypothesis Development:** Clearly defined variables help in formulating testable hypotheses.
- **Experimental Design:** Knowledge of how to manipulate and measure variables is crucial for designing effective experiments.
- **Data Analysis:** Identifying variable types informs appropriate statistical analyses and interpretation of results.

Research Hypothesis

- The research hypothesis is a predictive statement that relates an independent variable to dependent variable.
- A hypothesis may be defined as a proposition or set of proposition set forth as an explanation for occurrence of some specified group of phenomena either asserted merely as a provisional conjecture to guide some investigation or accepted as highly probable in the light of established facts

Purpose of hypothesis

- Guides/gives direction to the study/investigation
- Defines Facts that are relevant and not relevant
- Suggests which form of research design is likely to be the most appropriate
- Provides a framework for organizing the conclusions of the findings
- Limits the research to specific area
- Offers explanations for the relationships between those variables that can be empirically tested
- Furnishes proof that the researcher has sufficient background knowledge to enable her/him to make suggestions in order to extend existing knowledge
- Structures the next phase in the investigation and therefore furnishes continuity to the examination of the problem

Types of hypothesis

- **Based on formulation:** Null and Alternate hypothesis
- **Based on direction:** Directional and non directional hypothesis

Null and Alternative Hypotheses

- **Null Hypothesis (H_0):** The null hypothesis is a statement asserting that there is no effect, no difference, or no relationship between variables in a population. It serves as a default position that researchers aim to test against.
- **Example:** In a study examining the effect of a new drug on blood pressure, the null hypothesis might state that the mean blood pressure of patients taking the drug is equal to that of those not taking it ($H_0: \mu_1 = \mu_2$).
- **Alternative Hypothesis (H_1 or H_a):** The alternative hypothesis is a statement that contradicts the null hypothesis, suggesting that there is an effect, a difference, or a relationship present. It represents what the researcher aims to support through evidence.
- **Example:** Continuing with the drug study, the alternative hypothesis could state that the mean blood pressure of patients taking the drug is different from that of those not taking it ($H_1: \mu_1 \neq \mu_2$).

Directional and Non-Directional Hypotheses

- **Directional Hypothesis:** A directional hypothesis specifies the expected direction of the relationship or effect between variables. It indicates whether one variable is expected to be greater than or less than another.
- **Example:** In the context of the drug study, a directional hypothesis might state that patients taking the new drug will have lower blood pressure than those not taking it ($H_1: \mu_1 < \mu_2$).
- **Non-Directional Hypothesis:** A non-directional hypothesis does not specify a direction; it simply states that there is a difference or effect without indicating whether it is positive or negative.
- **Example:** Using the same drug study, a non-directional hypothesis would assert that there is a difference in mean blood pressure between patients taking the drug and those not taking it ($H_1: \mu_1 \neq \mu_2$).

Importance of hypothesis in Research

Understanding these hypotheses is crucial for conducting statistical tests and interpreting research findings effectively.

- **Null Hypothesis (H_0):** Assumes no effect or relationship exists.
- **Alternative Hypothesis (H_1):** Suggests an effect or relationship exists.
- **Directional Hypothesis:** Predicts the direction of the effect (greater than or less than).
- **Non-Directional Hypothesis:** Indicates there is a difference but does not specify direction.

Treatment

Treatment is what is administered to participants or experimental units in an experiment. It can take various forms, such as a drug, therapy, educational program, or any other intervention that is hypothesized to have an effect.

The different conditions under which experimental and control groups are put are usually referred to as 'treatments'

Experimental and Control Groups

Experimental and control groups are foundational elements in research design. The experimental group allows researchers to test hypotheses by applying treatments, while the control group provides a reference point for evaluating those effects.

Together, they enable rigorous testing of theories and contribute to reliable scientific conclusions.

In an experimental hypothesis-testing research when a group is exposed to usual conditions, it is termed a 'control group', but when the group is exposed to some novel or special condition, it is termed an 'experimental group'. In the above illustration, the Group A can be called a control group and the Group B an experimental group. If both groups A and B are exposed to special studies programmes, then both groups would be termed 'experimental groups.'

Does a new drug reduce blood pressure?

- **Experimental Variable (IV):** Type of drug given (new drug vs placebo).
- **Dependent Variable (DV):** Patient's blood pressure.
- **Control Variables:** Patient's diet, exercise routine, room temperature during testing, measurement method.

Does dataset size improve AI model accuracy?

- **Experimental Variable (IV):** Dataset size (10k, 50k, 100k images).
- **Dependent Variable (DV):** Model accuracy.
- **Control Variables:** Same model architecture, same optimizer, same learning rate, same evaluation dataset

Experimental Group

The experimental group is the group of subjects that receives the treatment or intervention being tested. This group is exposed to the independent variable to observe its effects on the dependent variable.

Characteristics:

- **Treatment Application:** Members of the experimental group experience changes or interventions that are hypothesized to affect outcomes.
- **Comparison Basis:** The results from this group are compared against those from the control group to determine the effectiveness of the treatment.

Example: In a clinical trial testing a new medication, the experimental group would receive the actual drug being studied.

Control Group

The control group is a separate group that does not receive the treatment or intervention. Instead, it serves as a baseline to compare against the experimental group.

Characteristics:

- **No Treatment or Placebo:** Members of the control group may receive a placebo or standard treatment, allowing researchers to isolate the effects of the independent variable.
- **Similarity to Experimental Group:** Ideally, control and experimental groups should be identical in every way except for the treatment being tested. This ensures that any differences in outcomes can be attributed to the treatment.

Example: In clinical trial, the control group might receive a placebo pill that has no therapeutic effect.

Importance of Experimental and Control Groups

- **Establishing Causality:** By comparing outcomes between groups, researchers can determine whether changes in the dependent variable are due to the independent variable.
- **Reducing Bias and Variability:** Control groups help mitigate confounding variables and biases, enhancing the validity of findings.
- **Benchmarking Results:** The control group provides a standard against which researchers can measure changes observed in the experimental group.

Types of Treatments

- **Experimental Treatments:** These are the new or modified interventions being tested (e.g., a new medication).
- **Control Treatments:** These may include standard treatments, placebos, or no treatment at all, used for comparison against the experimental treatments.

Purpose

The primary goal of applying treatments is to determine their effect on the outcome of interest (the dependent variable). By comparing the results from treatment groups to control groups, researchers can assess whether the treatment has a significant impact.

Design Considerations:

- Researchers must carefully design the treatment conditions, including dosage, duration, and delivery method, to ensure valid and reliable results.
- Random assignment of treatments helps minimize bias and confounding variables, enhancing the internal validity of the study.

Measurement:

- Outcomes are typically measured before and after treatment application (pretest and posttest) to evaluate changes attributable to the treatment.

Example of Treatment in Research

In a clinical trial evaluating the effectiveness of a new drug for hypertension:

- **Treatment Group:** Participants receive the new drug.
- **Control Group:** Participants receive a placebo.

Researchers compare blood pressure readings before and after treatment in both groups to determine if the new drug is effective.

Experiment

- The process of examining the truth of a statistical hypothesis, relating to some research problem, is known as an experiment.
- Experiments can be of two types viz., absolute experiment and comparative experiment.
- If we want to determine the impact of a fertilizer on the yield of a crop, it is a case of absolute experiment; but if we want to determine the impact of one fertilizer as compared to the impact of some other fertilizer, our experiment then will be termed as a comparative experiment. Often, we undertake comparative experiments when we talk of designs of experiments.

Experiment Units

- The pre-determined plots or the blocks, where different treatments are used, are known as experimental units. Such experimental units must be selected (defined) very carefully.

Experimental Design

Experimental design methods allow the experimenter to understand better and evaluate the factors that influence a particular system by means of statistical approaches.

Such approaches combine theoretical knowledge of experimental designs and a working knowledge of the particular factors to be studied.

3 principles of Experimental Design:

Principle of Randomization

- Design an experiment in such a way that all variations caused by extraneous factors are combined under general heading of “chance”.
- Assign variety of rice to be grown in different parts of field on basis of random sampling. This gives better estimate of experimental error.

Ex: Suppose a researcher wants to study the effect of three fertilizers (A, B, C) on crop yield. The field has different soil fertility levels — so we treat each fertility level as a block.

In each block:

- Fertilizers (treatments) are assigned randomly to plots.
- This controls the soil variation effect.

Principle of Replication

- Experiment should be repeated more than once.
- Each treatment is applied in many experimental units instead of one. Hence, statistical accuracy is increased.
- Eg: Divide the field into several parts, grow one variety in one half of these parts and another variety in another half of the parts. Collect the data of yield of the two varieties and draw conclusion by comparing the same. The result is more reliable than that without replication.

Principle of Local Control

- Divide the field into several homogenous parts called block and divide block into several parts (equal to number of treatments).
- Blocks are the levels at which we hold extraneous factor fixed so that we can measure its contribution to total variability of data.
- It eliminates variability due to extraneous factor from experimental error.
- Eg: Treatment (Different rice varieties); Extraneous factor (soil fertility)

1. Randomization

Meaning: Randomly assign subjects or experimental units to different treatments.

Purpose: Eliminates bias and spreads unknown variables evenly across groups.

Example: Randomly assign students to different teaching methods.

2. Replication

Meaning: Repeat the experiment or apply the treatment to multiple subjects.

Purpose: Increases reliability and allows estimation of experimental error.

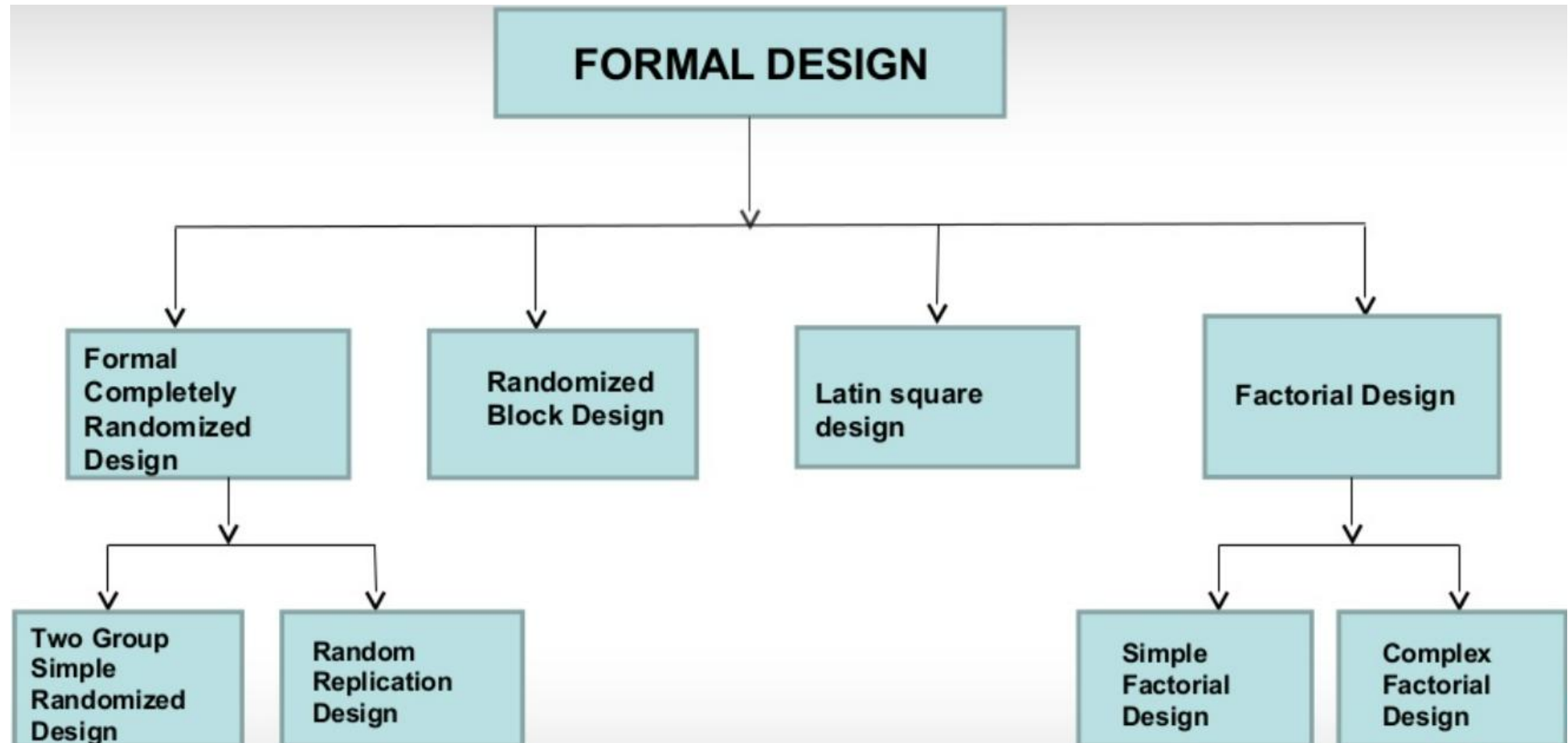
Example: Use the same fertilizer on several plots to confirm consistent results.

3. Local Control

Meaning: Divide experimental units into homogeneous blocks (groups) based on similar characteristics.

Purpose: Reduces variability caused by external factors.

Example: Group fields by soil type before applying fertilizers.



Single factor/Single independent variable

Multi factors/Multiple independent variables

Formal design is a predetermined experimental plan that specifies how treatments are assigned to experimental units and how the resulting observations will be analyzed.

For example, if you want to find out whether three different teaching methods affect students' marks, you need to decide:

How many students? → Which teaching method? → How to assign students? → How to collect marks? → How to compare the groups?

That plan is the experimental design.

The terms **experimental design** and **formal design** are related, but they are not exactly the same. A simple way to understand them is:

- Formal design is a structured plan for conducting research, while experimental design is a specific type of formal design used when the researcher conducts an experiment.

Simple example

Suppose you want to study whether fertilizer affects plant growth.

- **Formal research design:**

- Select the plants.
- Decide what data to collect.
- Decide how many plants to use.
- Decide when to collect measurements.
- Decide how to analyze the data.
- This gives the overall structured plan.

- **Experimental design:**

- Fertilizer A → Group 1, Fertilizer B → Group 2
- No fertilizer → Control group
- Randomly assign plants to groups, Keep water, soil, sunlight, etc. controlled.
- Measure plant growth.
- Compare the results statistically.

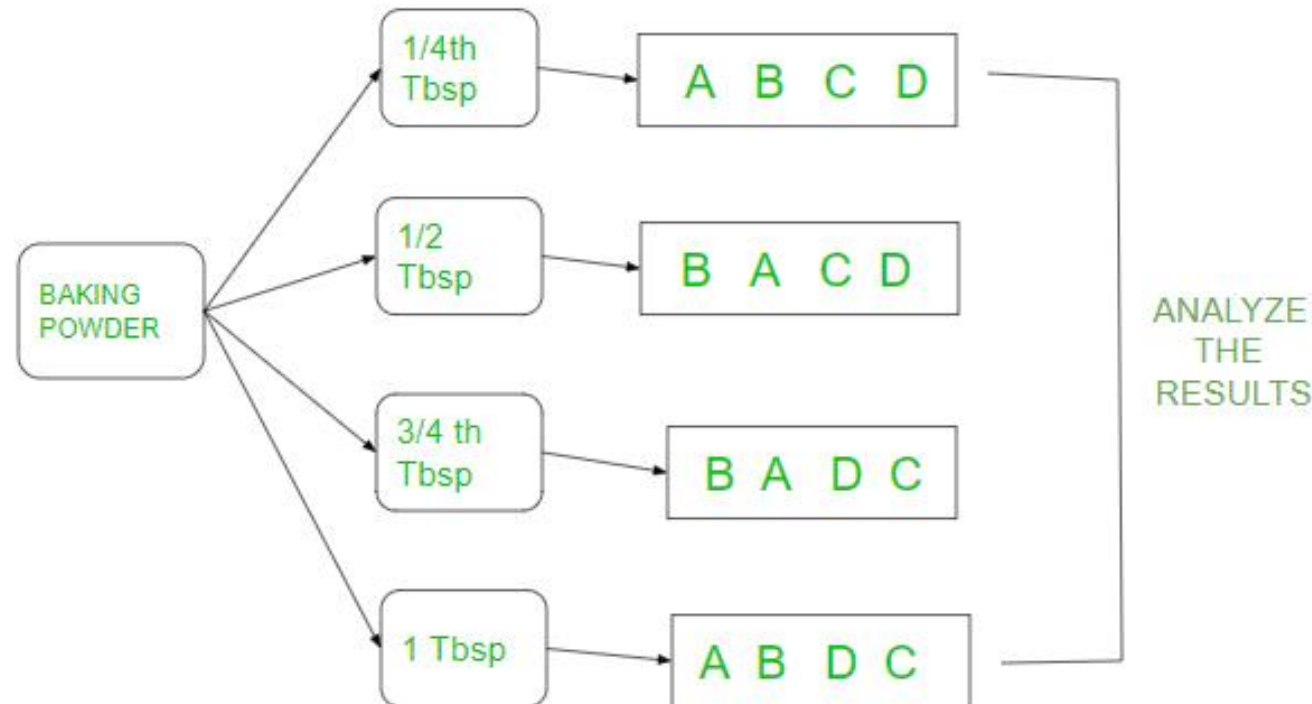
Completely Randomised Design

Completely Randomized Design (CRD) is a basic experimental design in which treatments are assigned to experimental units completely at random. This means that each unit has an equal chance of receiving any treatment, ensuring that the effects of the treatments can be assessed without bias.

Applies principles of replication and randomization

□ Example

- Adding more baking powder to cakes increases the heights of the cake



- **Key Characteristics**

- 1.Random Assignment:** Treatments are allocated randomly to experimental units, which minimizes the influence of uncontrolled variables and helps in achieving homogeneity among groups.
- 2.Single Factor Focus:** CRD is typically used when studying the effects of one primary factor (treatment) on a response variable.
- 3.Homogeneity Requirement:** It is most effective when the experimental units are homogeneous, meaning they are similar in all respects except for the treatment being tested.
- 4.Flexibility:** The design allows for any number of treatments and replications, making it adaptable to various experimental conditions.

Statistical Model

The statistical model for CRD can be expressed as:

$$Y_{ij} = \mu + T_i + e_{ij}$$

Where:

- Y_{ij} = observation for treatment i and replication j
- μ = overall mean
- T_i = effect of treatment i
- e_{ij} = random error associated with the observation

- **Advantages**

- **Simplicity:** The layout and analysis of CRD are straightforward, making it easy to implement.
- **Statistical Efficiency:** It provides maximum degrees of freedom for error, which enhances the sensitivity of statistical tests.
- **Flexibility in Design:** Researchers can easily modify treatment levels and replications as needed.

- **Limitations**

- **Homogeneity Requirement:** CRD is less effective if the experimental units are not homogeneous, as this can lead to increased variability and reduced precision.
- **Less Accurate in Field Experiments:** Due to environmental variations, CRD may not be suitable for field experiments where conditions are not controlled.
- **Inability to Detect Interactions:** It does not account for interactions between factors, which may limit its applicability in more complex experiments.

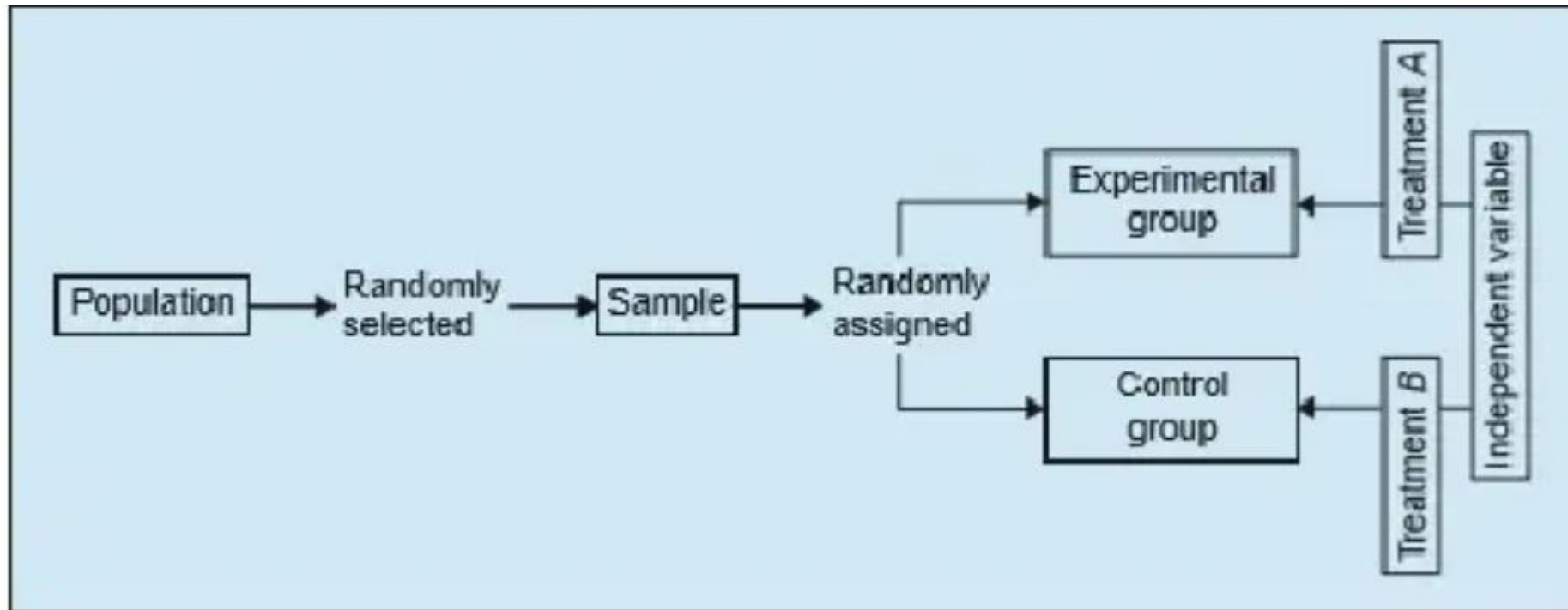
□ Applications

- CRD is widely used in laboratory experiments where conditions can be tightly controlled, such as: Agricultural research to test different treatments on crops.
- Clinical trials to assess the effectiveness of new drugs.
- Basic research in biology and chemistry where randomization is crucial for unbiased results.

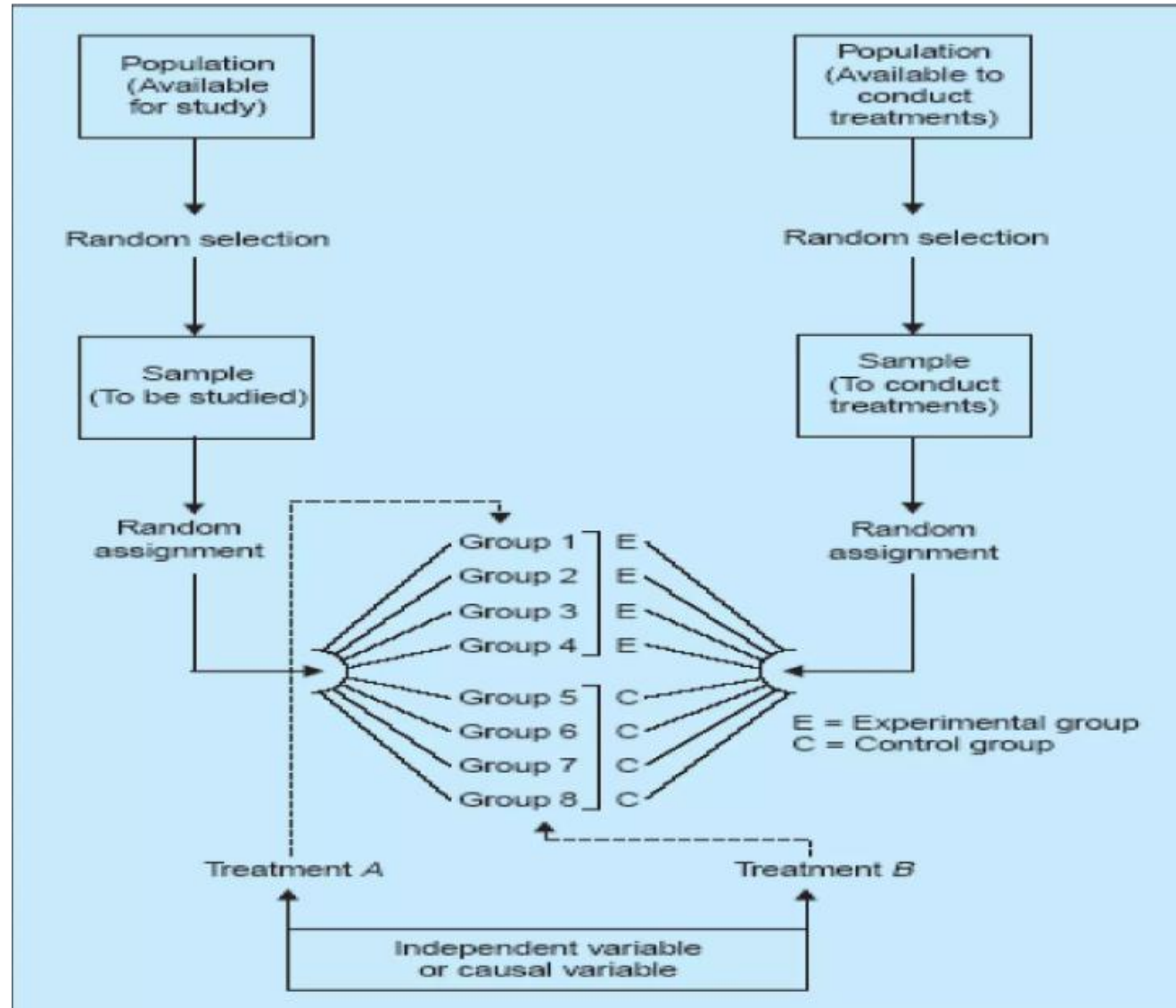
Types

- **Two group simple randomized experimental design:** Population is defined and samples are selected. It is randomly assigned to experimental and control groups. The two groups are given different treatment of independent variable. Suitable behavioral science research. Eg: Normal vs specialized training on students.
- **Random replications design:** Repetition for each treatment. It provides controls for differential effects of extraneous independent variables and secondly, it randomizes individual differences among those conducting the experiments.

Two group simple randomized experimental design



Random replications design

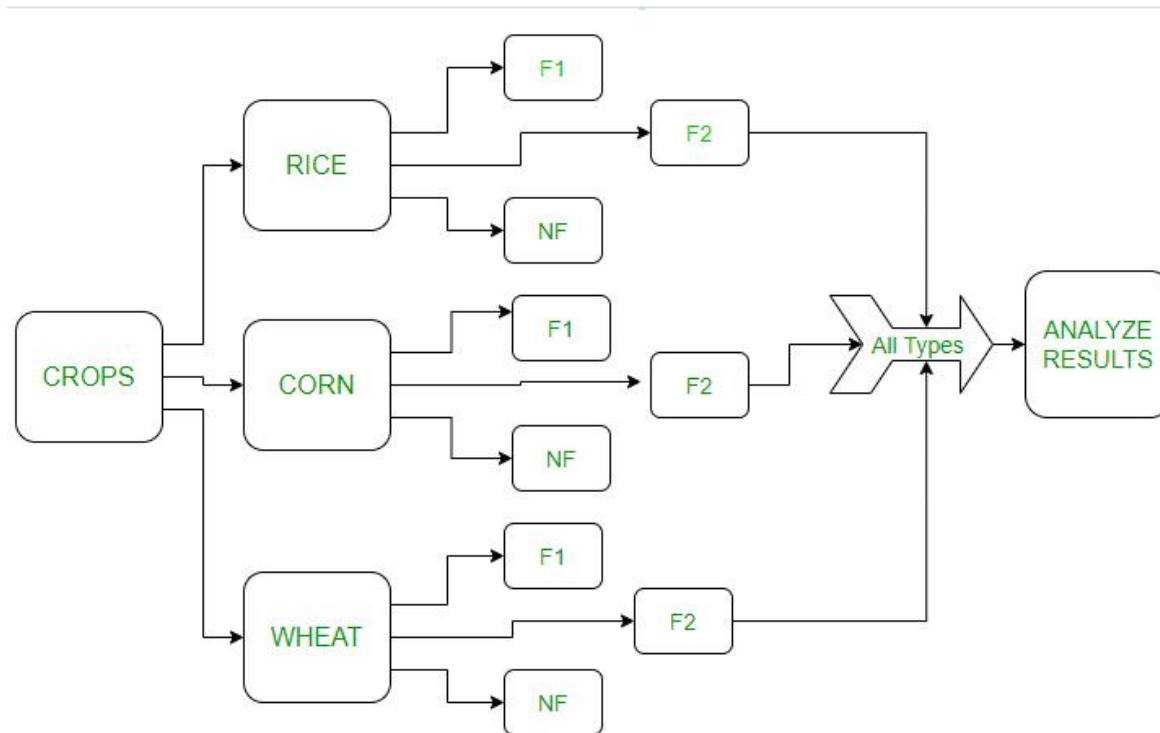


Randomized Block Design

Randomized Block Design (RBD) is an experimental design that involves grouping experimental units into blocks based on certain characteristics or factors that are expected to affect the outcome of the experiment. Within each block, treatments are assigned randomly to minimize the impact of variability between blocks.

□ Example

- Testing new fertilizers in different types of crops. Crops are divided into 3 different types(blocks).These blocks are again divided into 3 fertilizers that are used on those crops



- **Key Features**

- 1.Blocking:** Experimental units are grouped into blocks that share similar characteristics (e.g., age, gender, location). This helps control for variability and reduces the experimental error associated with those nuisance factors.
- 2.Random Assignment:** Treatments are randomly assigned within each block, ensuring that each treatment has an equal chance of being applied to any experimental unit in the block.
- 3.Complete Treatment Application:** Each treatment is applied to all blocks, allowing for a comprehensive comparison across different conditions.

Statistical Model

The statistical model for RBD can be expressed as:

$$Y_{ij} = \mu + T_j + B_i + E_{ij}$$

Where:

- Y_{ij} = observation for treatment j in block i
- μ = overall mean
- T_j = effect of treatment j
- B_i = effect of block i
- E_{ij} = random error associated with the observation

- **Advantages**

- 1.Reduction of Variability: By accounting for known sources of variability, RBD increases the precision of the experiment and enhances the reliability of results.
- 2.Increased Statistical Power: The design allows for more accurate estimates of treatment effects by reducing the error variance, leading to more powerful statistical tests.
- 3.Flexibility: RBD can accommodate various types of experimental conditions and is adaptable to many fields, including agriculture, medicine, and social sciences.

- **Limitations**

- 1.Complexity in Design: Designing an experiment with appropriate blocking factors can be complex and requires prior knowledge about potential sources of variability.
- 2.Assumption of Homogeneity: The effectiveness of blocking relies on the assumption that variability within blocks is less than between blocks. If this assumption is violated, it may lead to misleading results.
- 3.Limited to Known Factors: RBD only accounts for known nuisance factors; unknown sources of variability may still affect the results.

□ Applications

- Agricultural Experiments: Used to test different fertilizers or treatments across fields with varying soil types or environmental conditions.
- Clinical Trials: Employed to assess treatment effects while controlling for patient demographics or baseline characteristics.
- Industrial Testing: Applied in quality control processes where variations in production conditions need to be accounted for.

Design	Main idea	Example
CRD	Randomly assign treatments	Compare 3 fertilizers on similar plants
RBD	Create blocks of similar units, then randomize	Compare fertilizers while controlling soil quality
Latin Square	Control two sources of variation	Compare fertilizers while controlling soil type and location
Factorial	Study two or more factors simultaneously	Study teaching method + study material

Latin Square Design

Latin Square Design (LSD) is an experimental design used to control for two blocking factors while studying the effects of one treatment factor. It is structured as an $n \times n$ grid where each treatment appears exactly once in each row and once in each column, effectively controlling for variability due to the two blocking factors.

For example, suppose we want to compare **4 teaching methods**, but students differ according to:

- Ability level
- Time/session

A Latin square can control both factors.

- For 4 treatments:

	Column 1	Column 2	Column 3	Column 4
Row 1	A	B	C	D
Row 2	B	C	D	A
Row 3	C	D	A	B
Row 4	D	A	B	C

Selected Latin Squares

3 x 3

A B C A B C D

B C A B A D C

C A B C D B A

 D C A B

4 x 4

A B C D

B C D A

C D A B

D A B C

A B C D

B D A C

C A D B

D C B A

A B C D

B A D C

C D A B

D C B A

5 x 5

A B C D E

B A E C D

C D A E B

D E B A C

E C D B A

6 x 6

A B C D E F

B F D C A E

C D E F B A

D A F E C B

E C A B F D

F E B A D C

Example:

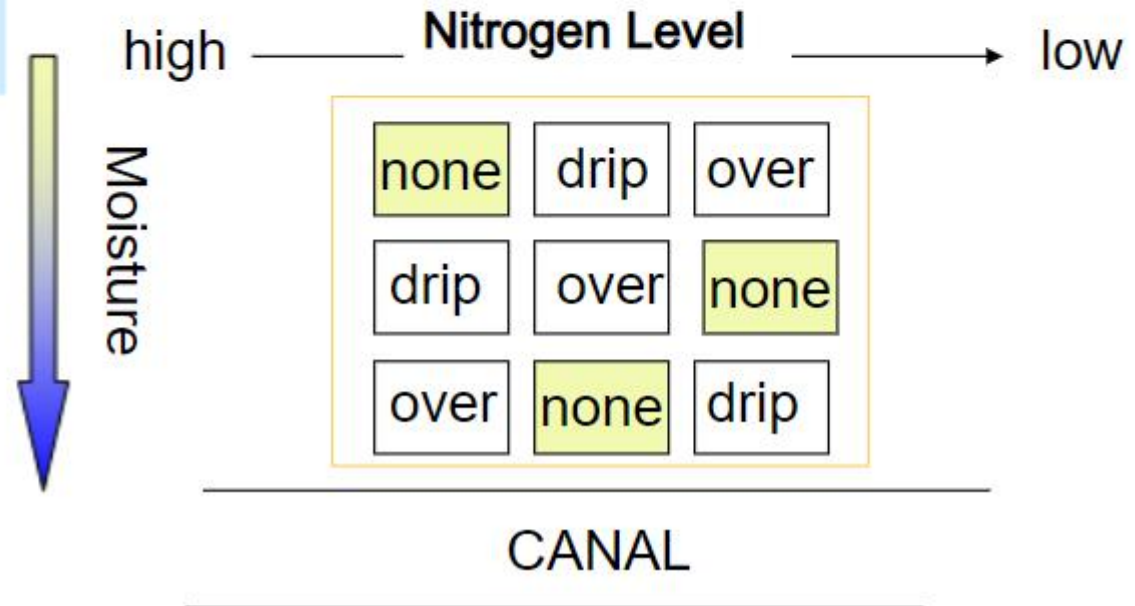
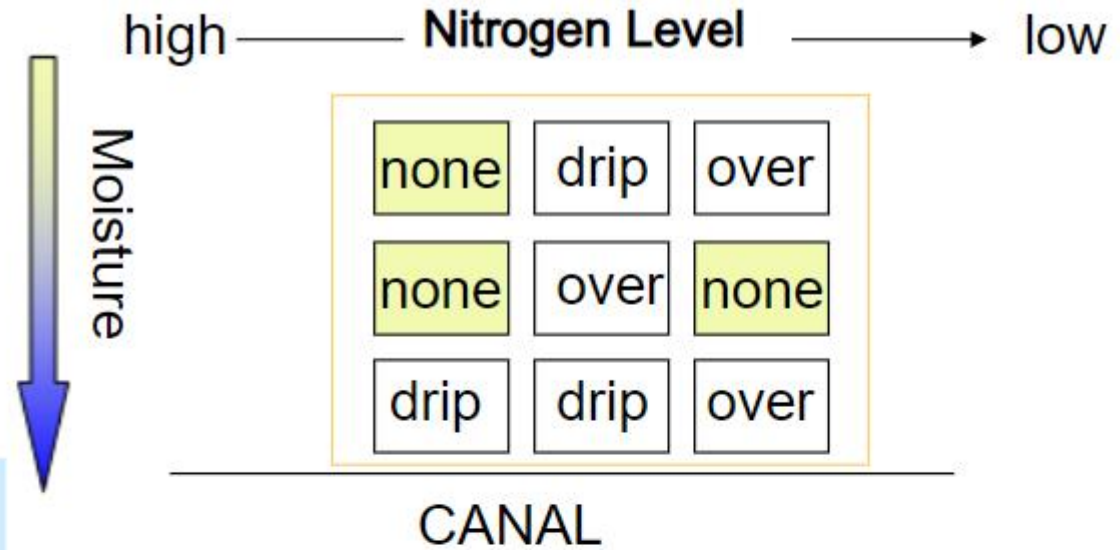
Cold Protection of Strawberries

- Three different irrigation methods (**treatment levels**) are used on strawberries:
 1. drip,
 2. overhead sprinkler,
 3. no irrigation.
- We wish to determine which of these is most effective in protecting strawberries from extreme cold.
- All strawberries grown through plastic mulch.
- Measure weight of frozen fruit (lower values indicate more protection).

Field Layout

Moisture and Soil Nitrogen are two sources of extraneous variation that we wish to simultaneously control for.

Which design will best allow us to account for both soil moisture and nitrogen gradients?



- **Key Features**

- 1.Structure:** The design consists of a square matrix where both rows and columns represent different blocking factors. Each cell in the matrix contains a treatment, ensuring that every treatment is tested under every combination of the blocking factors.
- 2.Blocking Factors:** The two blocking factors are usually known nuisance variables that could affect the outcome of the experiment. By arranging treatments in this way, researchers can isolate the effects of the treatment from those of the blocking factors.
- 3.Equal Treatment Replication:** Each treatment appears once in each row and column, allowing for balanced comparisons across all conditions.

Statistical Model

The statistical model for analyzing data from a Latin Square Design can be represented as:

$$Y_{ijk} = \mu + T_j + R_i + C_k + E_{ijk}$$

Where:

- Y_{ijk} = observation for treatment j in row i and column k
- μ = overall mean
- T_j = effect of treatment j
- R_i = effect of row i
- C_k = effect of column k
- E_{ijk} = random error associated with the observation

- **Advantages**

1. **Control of Variability:** By accounting for two sources of variability (rows and columns), LSD reduces experimental error and increases the precision of results.
2. **Efficient Use of Resources:** The design requires fewer experimental units compared to running separate experiments for each blocking factor.
3. **Flexibility:** It can accommodate various types of treatments and is applicable in multiple fields, including agriculture, psychology, and clinical studies.

- **Limitations**

1. **Limited Number of Treatments:** The number of treatments must equal the number of rows and columns, which can limit its applicability if there are too many treatments.
2. **Complexity in Design:** Designing a Latin square can be more complex than simpler designs, requiring careful planning to ensure that all treatments are adequately represented.
3. **Assumption of Homogeneity:** The effectiveness relies on the assumption that variations within rows and columns are minimal; significant differences may lead to biased results.

□ Applications

- Agricultural Experiments: Used to evaluate different fertilizers or crop treatments while controlling for soil type (rows) and environmental conditions (columns).
- Clinical Trials: Employed to study drug effects while accounting for patient demographics (rows) and treatment regimens (columns).
- Psychological Studies: Applied in experiments where subjects may differ based on characteristics like age or gender, ensuring balanced representation across treatments.

Factorial Design

Factorial design is a statistical experimental methodology used to evaluate the effects of two or more independent variables (factors) on a dependent variable. This design allows researchers to assess not only the main effects of each factor but also the interaction effects between factors, providing a comprehensive understanding of how these variables influence outcomes.

- **Key Features**

1. Multiple Factors: Factorial designs involve at least two independent variables, each with multiple levels. For example, a 2x2 factorial design includes two factors, each at two levels, resulting in four treatment combinations.

2. Main and Interaction Effects:

Main Effects: The individual impact of each factor on the dependent variable.

Interaction Effects: How the effect of one factor changes depending on the level of another factor.

3. Comprehensive Analysis: By manipulating multiple factors simultaneously, factorial designs allow for a nuanced analysis that can reveal complex relationships that might not be evident when studying factors in isolation.

Example: Teaching experiment

Suppose we want to study:

Factor A: Teaching method

- Traditional
- Online

Factor B: Study material

- Textbook
- Video

This gives a **2 × 2 factorial design**:

	Textbook	Video
Traditional teaching	Group 1	Group 2
Online teaching	Group 3	Group 4

From the above example, We can analyze:

Effect of teaching method

Does online/traditional teaching affect marks?

Effect of study material

Does textbook/video material affect marks?

Interaction

Does the effectiveness of the teaching method depend on the type of study material?

This interaction is one of the major advantages of factorial designs.

Simple Factorial Design

A simple factorial design typically involves two or more factors, each with two levels. It is often represented in a format like 2x2, where the first number indicates the number of levels for the first factor and the second number indicates the levels for the second factor.

Example: In a 2x2 factorial design examining the effects of drug dosage (low vs. high) and treatment duration (short vs. long), there would be four treatment combinations:

- Low dosage, short duration
- Low dosage, long duration
- High dosage, short duration
- High dosage, long duration

Complex Factorial Design

Complex factorial designs involve more than two factors or factors with more than two levels. They can be represented as 2×3 or $3 \times 2 \times 2$, indicating multiple factors and their respective levels.

Example: In a $3 \times 2 \times 2$ factorial design examining the effects of three factors—type of therapy (Cognitive Behavioral Therapy, Psychodynamic Therapy, and Humanistic Therapy), duration (short vs. long), and therapist gender (male vs. female)—there would be 12 treatment combinations:

- CBT, short duration, male
- CBT, short duration, female
- CBT, long duration, male
- CBT, long duration, female
- Psychodynamic, short duration, male
- Psychodynamic, short duration, female
- Psychodynamic, long duration, male
- Psychodynamic, long duration, female
- Humanistic, short duration, male
- Humanistic, short duration, female
- Humanistic, long duration, male
- Humanistic, long duration, female

Advantages of factorial design

➤ There are many advantages of the factorial experiments:

1. Factorial experiments are advantageous to study the combined effect of two or more factors simultaneously and analyze their interrelationships. Such factorial experiments are economic in nature and provide a lot of relevant information about the phenomenon under study. It also increases the efficiency of the experiment.
2. It is an advantageous because a wide range of factor combination are used. This will give us an idea to predict about what will happen when two or more factors are used in combination.
3. The factorial approach will result in considerable saving of the time and the experimental materials. It is because the time required for the combined experiment is less than that required for the separate experiments.
4. In single factor experiments, the results may not be satisfactory because of the changes in environmental conditions. However, in factorial experiments such type of difficulties will not arise even after several factors are investigated simultaneously.
5. Information may obtained from factorial experiments is more complete than that obtained from a series of single factor experiments, in the sense that factorial experiments permit the evaluation of interaction effects.

Disadvantages of factorial design

➤ The disadvantages of the factorial experiments are:

- It is disadvantageous because the execution of the experiment and the statistical analysis becomes more complex when several treatments combinations or factors are involved simultaneously.
- It is also disadvantageous in cases where may not be interested in certain treatment combinations but we are forced to include them in the experiment. This will lead to wastage of time and also the experimental material.
- In factorial experiments, the number of treatment combinations will increase if the factors are increased. This will also lead to the increase in block size, which in turn will increase the heterogeneity in the experimental material. Because of this it will lead to the increased experimental error and will decrease the precision in the experiment. Appropriate block size must be maintained.

Comparison between formal & experimental design w.r.t following example.

Student Performance Prediction using ML: Suppose the project objective is:

“Predict whether a student will pass or fail based on attendance, internal marks, study hours and assignment performance”.

Formal design component

Research problem

Objective

Input variables

Output variable

Data source

Sampling

Data preprocessing

Training data

Testing data

Methods/treatments

Performance measures

Comparison

Conclusion

ML project equivalent

Predict student pass/fail

Develop a model to predict student performance

Attendance, study hours, internal marks, assignment marks

Pass / Fail

Student historical records

Select a representative set of student records

Missing values, normalization, encoding

80% of dataset

20% of dataset

Decision Tree, Random Forest, SVM, KNN

Accuracy, Precision, Recall, F1-score

Compare the performance of different algorithms

Determine which algorithm meets the predefined performance criteria

How does this become a "formal design"?

The researcher decides these things before carrying out the analysis.

But where does "**experimental design**" come in?

Suppose you are studying:

"Effect of different ML algorithms on waste classification accuracy."

You can consider:

Independent variable: ML algorithm

Dependent variable: Classification accuracy

Experimental conditions: Same dataset, same train-test split, same preprocessing

Treatments: Decision Tree, Random Forest, SVM, CNN

Replications: Repeat the experiment using different random seeds or cross-validation folds

Response: Accuracy/F1-score

Formal design

The **overall plan** of the ML project:

Dataset → preprocessing → feature selection → model development → training → testing → evaluation → comparison.

Experimental design

The **controlled comparison** within that project:

Decision Tree vs Random Forest vs SVM under the same experimental conditions.

Thank You